Research statement

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Research focus

I investigate battery management systems where physical models, online estimation and intelligent computation cooperate without making immediate safety dependent on remote services. Lithium-ion batteries require internal-state estimates even though important states cannot be measured directly. Cloud-based digital twins may improve analysis, but they also introduce communication, latency and availability constraints.

Central research question

How can a battery-management system request additional cloud intelligence only when it is useful, while keeping local estimation and protection operational during communication faults or cloud outages?

Present contribution

For my current research I developed a safety-gated, event-triggered edge-cloud digital-twin architecture around a four-cell lithium-ion battery system. Each cell retains distinct state information in a reduced-order equivalent-circuit and thermal representation. The local BMS includes an EKF-type state estimator, residual and uncertainty monitoring, safety supervision, current derating, balancing, and contactor / pre-charge logic. Communication requests depend on operating-risk indicators. Returned cloud corrections are checked for state consistency, freshness, confidence and physical feasibility before affecting local estimation.

Measured result and its boundaries

In the reported tested Processor-in-the-Loop (PIL) configuration, the event-triggered strategy reduced edge-to-cloud communication by approximately 82.8% relative to continuous communication. The corresponding 17.2% normalized communication activity is a derived comparison, not an additional measured trace. This result is configuration-specific; it does not establish theoretical estimator stability or universal behaviour under all operating conditions. The supplied report presents HPPC and UDDS current, voltage, temperature and reference-SOC plots at 25°C, which do not by themselves constitute estimator-error comparisons.

Source: Scientific project report, October 2026, Sections 4-6, 14, 15 and 23.

Validation and future research

Model and evaluation approach

The local model relates terminal voltage to open-circuit voltage, ohmic drop and polarisation dynamics. Residuals are defined by the difference between measured and predicted signals and help determine whether an update is useful. The available work includes simulation and PIL-oriented validation, including the reported communication comparison. The next stage is physical Controller Hardware-in-the-Loop (CHIL) testing with an OPAL-RT real-time plant and a TI C2000 embedded controller.

What still needs experimental proof

The next CHIL campaign is intended to examine deterministic execution, end-to-end latency, sensor faults, balancing behaviour, contactor and pre-charge operation, delayed packets and cloud outage recovery. No completed physical CHIL timing measurements are claimed here. Some trigger and safety-gate parameters remain engineering thresholds rather than formally derived bounds. Long-term SOH / RUL prognosis and detailed physics-informed models are not yet established within the fast safety-critical loop.

Proposed PhD research directions

I would like to build on this work in three directions: (1) connecting event-trigger thresholds and estimator uncertainty to analyzable communication-performance trade-offs; (2) modelling cloud feedback as a verifiable contract with explicit validity requirements; and (3) exploring physics-informed state and ageing models that can transfer across cell conditions while remaining compatible with resource-constrained embedded implementation. These are future scientific questions, not completed results.

Working approach and fit

My background in mechanical engineering, energy systems, computer vision and battery-related ML motivates an approach that connects physical assumptions to numerical modelling, reproducible evaluation and real-time validation. I am interested in laboratories working on battery state estimation, networked control, formal verification of cyber-physical systems, and model-informed intelligent energy systems.

Traceability

The research content in this statement comes from the supplied October 2026 project report and uploaded academic CV. The complete working report and private battery project repository are not redistributed with the public portfolio by default. This statement is a research summary; it is not a journal publication.

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